Further Trigonometry
Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

1 a. Sketch the curve $y = sin^{-1}\frac{4x-1}{2}$. 2

Domain: $-1 \le \frac{4x-1}{2} \le 1$

$-2 \le 4x-1 \le 2$

$-1 \le 4x \le 3$

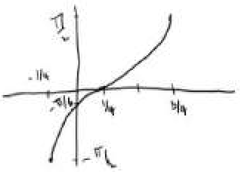
$-\frac{1}{4} \le x \le \frac{3}{4}$

Range: $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$

$(x=0)$: $y = \sin^{-1}\left(-\frac{1}{2}\right)$

$= -\sin^{-1}\left(\frac{1}{2}\right)$

$= -\frac{\pi}{6}$



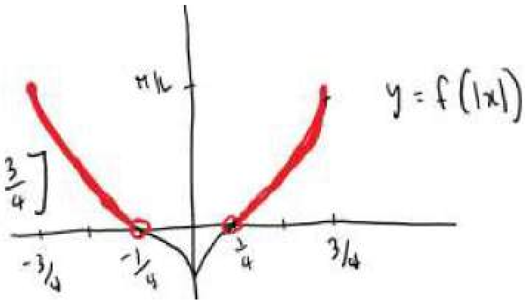
[2]

b. Hence or otherwise, solve $sin^{-1}\frac{4|x|-1}{2} > 0$. 1

$\therefore f(|x|) = \sin^{-1}\frac{4|x|-1}{2}$

$f(|x|) > 0$

$\Rightarrow x \in \left[-\frac{3}{4}, \frac{1}{4}\right) \cup \left(\frac{1}{4}, \frac{3}{4}\right]$



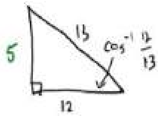
2 Find the exact value of $tan\left(2\,cos^{-1}\frac{12}{13}\right)$. 2

$\tan\left(2\cos^{-1}\frac{12}{13}\right) = \frac{2\tan\left(\cos^{-1}\frac{12}{13}\right)}{1-\tan^2\left(\cos^{-1}\frac{12}{13}\right)}$

$= \frac{2\left(\frac{5}{12}\right)}{1-\left(\frac{5}{12}\right)^2} \times \frac{144}{144}$

$= \frac{120}{144-25}$

$= \boxed{\frac{120}{119}}$



[2]

3 What is the value of $cos^{-1}\left(-\frac{1}{2}\right)$? 1

$\cos^{-1}\left(-\frac{1}{2}\right) = \pi - \cos^{-1}\frac{1}{2} = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$

4 a. Use $t-$formulae to show that $\frac{1-cos\theta}{sin\theta} = tan\left(\frac{\theta}{2}\right)$. 1

$$\text{LHS} = \frac{1-\cos\theta}{\sin\theta} = \frac{1-\left(\frac{1-t^2}{1+t^2}\right)}{\frac{2t}{1+t^2}}$$

$$= \frac{1+t^2-1+t^2}{2t} = \frac{2t^2}{2t} = t = \text{RHS}$$

b. Hence, or otherwise, find the exact value of $tan 15^o$. 1

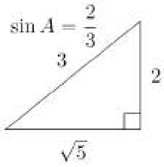
Let $\theta = 30^\circ$

$$\tan 15^\circ = \frac{1-\cos 30^\circ}{\sin 30^\circ} = \frac{1-\frac{\sqrt{3}}{2}}{\frac{1}{2}} = 2-\sqrt{3}$$

5 Find the exact value of $tan\left(2\,sin^{-1}\frac{2}{3}\right)$. 2

Let $A = \sin^{-1}\frac{2}{3}$

$\sin A = \frac{2}{3}$



$$\tan 2A = \frac{2\tan A}{1-\tan^2 A}$$

$$= \frac{2\times\frac{2}{\sqrt{5}}}{1-\frac{4}{5}}$$

$$= \frac{4}{\sqrt{5}}\times 5$$

$$= 4\sqrt{5}$$

6 What is the domain and range of $y = 2\,cos^{-1}(x-1)$? 1

$-1 \le x - 1 \le 1$ $\quad$ $0 \le cos^{-1} z \le \pi$

$\therefore 0 \le x \le 2$ $\quad$ $\therefore 0 \le 2 cos^{-1} z \le 2\pi$

$\therefore 0 \le y \le 2\pi$

7 By use of a suitable sketch, or otherwise, solve $sin^{-1}(3x+1) = cos^{-1} x$. 3

Find the domains and main points for the graph:

$\arcsin(-1) = -\frac{\pi}{2}$ $\quad$ When $3x+1 = -1$, $x = -\frac{2}{3}$

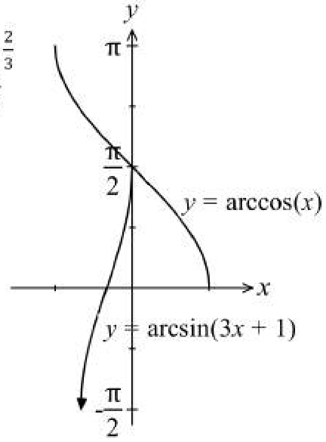
$\arcsin(0) = 0$ $\quad$ When $3x+1 = 0$, $x = -\frac{1}{3}$

$\arcsin(1) = \frac{\pi}{2}$ $\quad$ When $3x+1 = 1$, $x = 0$

$\arccos(-1) = \pi$

$\arcsin(0) = \frac{\pi}{2}$

$\arccos(1) = 0$ $\quad$ $\therefore x = 0.$



8 a. Prove the trigonometric Identity $tan3\theta = \frac{3tan\theta - tan^3\theta}{1-3\,tan^2\theta}$ using the compound angle formula. 2

$LHS = \tan 3\theta = \tan(2\theta + \theta)$

$= \dfrac{\tan 2\theta + \tan\theta}{1 - \tan 2\theta \tan\theta}$

$= \dfrac{\left(\frac{2\tan\theta}{1-\tan^2\theta}\right) + \tan\theta}{1 - \tan\theta \times \frac{2\tan\theta}{1-\tan^2\theta}}$

$= \dfrac{2\tan\theta + \tan\theta(1-\tan^2\theta)}{1-\tan^2\theta - 2\tan^2\theta}$

$= \dfrac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$

$= RHS$

3

b. Hence find expressions for the exact value of the solutions to the equation $\frac{3x-x^3}{1-3x^2} = \sqrt{3}$.

Let $\tan\theta = x$

$\therefore \dfrac{3x - x^3}{1-3x^2} = \sqrt{3}$

$\dfrac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta} = \sqrt{3}$

$\tan 3\theta = \sqrt{3}$

$3\theta = \frac{\pi}{3}, \frac{4\pi}{3}, \frac{7\pi}{3}$

$\theta = \frac{\pi}{9}, \frac{4\pi}{9}, \frac{7\pi}{9}$

$x = \tan\frac{\pi}{9}, \tan\frac{4\pi}{9}$, and $\tan\frac{7\pi}{9}$

2

c. Hence show that $tan\frac{\pi}{9} \times tan\frac{4\pi}{9} \times tan\frac{7\pi}{9} = -\sqrt{3}$.

$\dfrac{3x - x^3}{1-3x^2} = \sqrt{3}$

$\sqrt{3} - 3\sqrt{3}x^2 = 3x - x^3$

$x^3 - 3\sqrt{3}x^2 - 3x + \sqrt{3} = 0$

Since $\tan\frac{\pi}{9}$, $\tan\frac{4\pi}{9}$ and $\tan\frac{7\pi}{9}$ are the roots of this equation

$\tan\frac{\pi}{9} \times \tan\frac{4\pi}{9} \times \tan\frac{7\pi}{9} = -\frac{d}{a} = -\sqrt{3}$

9 Given $tan4\theta = \frac{4tan\theta - 4\,tan^3\,\theta}{1 - 6\,tan^2\,\theta + tan^4\,\theta}$, find the solutions to the equation $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$. 3

$$\tan 4\theta = \frac{4\tan \quad 4\tan^3}{1 \quad 6\tan^2 \quad +\tan^4} \qquad \text{If } x = \tan\theta, \text{ then } \tan 4\theta = \frac{4x \quad 4x^3}{1 \quad 6x^2 + x^4}$$

$$\frac{4x \quad 4x^3}{1 \quad 6x^2 + x^4} = 1 \quad \text{if } \tan 4\theta = 1$$

$$x^4 + 4x^3 \quad 6x^2 \quad 4x + 1 = 0 \text{ if } \tan 4\theta = 1, \quad \text{then } 4 \quad = n \quad + \frac{}{4} \text{ or } \quad = \frac{n}{4} + \frac{}{16}$$

Putting n =0,1,2 and 3 gives $= \frac{}{16}, \frac{5}{16}, \frac{9}{16}, \frac{13}{16}$; If the roots are x_1, x_2, x_3 and x_4

$$x_1 = \tan\frac{}{16}, x_2 = \tan\frac{5}{16}, x_3 = \tan\frac{9}{16}, x_4 = \tan\frac{13}{16}$$

10 Prove:

a. $tan(A + B + C) = \frac{tanA + tanB + tanC - tanAtanBtanC}{1 - tanAtanB - tanBtanC - tanCtanA}$. 2

$$tan(A+B) = \frac{tanA + tanB}{1 \quad tanAtanB}$$

$$\tan[A + (B + C)] = \frac{\tan A + \tan(B+C)}{1 \quad \tan A \tan(B+C)} = \frac{\tan A + \frac{\tan B + \tan C}{1 \quad \tan B \tan C}}{1 \quad \tan A \frac{\tan B + \tan C}{1 \quad \tan B \tan C}}$$

$$= \frac{\frac{\tan A \quad \tan A \tan B \tan C + \tan B + \tan C}{1 \quad \tan B \tan C}}{\frac{1 \quad \tan B \tan C \quad \tan A \tan B \quad \tan A \tan C}{1 \quad \tan B \tan C}} = \frac{\tan A + \tan B + \tan C \quad \tan A \tan B \tan C}{1 \quad \tan A \tan B \quad \tan B \tan C \quad \tan C \tan A}$$ 2

b. A, B and C are acute angles of a triangle and $\frac{tanA}{5} = \frac{tanB}{6} = \frac{tanC}{7} = k$. Show that $k = \sqrt{\frac{3}{35}}$ is the only solution to $tan(A + B + C) = 0$.

In ΔABC, $A+B+C = 180^0$

$\therefore \tan(A+B+C) = \tan 180^0 = 0$

$\therefore \tan A + \tan B + \tan C - \tan A \tan B \tan C = 0$

or $\tan A + \tan B + \tan C = \tan A \tan B \tan C$

Given $\frac{\tan A}{5} = \frac{\tan B}{6} = \frac{\tan C}{7} = k$

$\therefore \tan A = 5k$, $\tan B = 6k$ and $\tan C = 7k$

$5k + 6k + 7k = 5k \cdot 6k \cdot 7k$

$18k = 210\,k^3$ or $k(35k^2 - 3) = 0$

$\therefore k = 0 \quad k = \pm\sqrt{\frac{3}{35}}$ The only valid value of $k = +\sqrt{\frac{3}{35}}$ since $\tan A > 0$, $\tan B > 0$ and $\tan C > 0$ and A,B and C are acute.

11 Find the domain and range of $y = 4\,cos^{-1}\left(\frac{3x}{2}\right)$. 2

D: $-1 \le \frac{3x}{2} \le 1$

$-2 \le 3x \le 2$

$\frac{-2}{3} \le x \le \frac{2}{3}$

R: $0 \le \frac{y}{4} \le \pi$

$0 \le y \le 4\pi$

Further Trigonometry
Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

1 Evaluate $\lim_{\theta \to 0} \frac{2\sin\left(\pi + \frac{\theta}{2}\right)}{\theta}$. 1

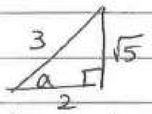
$$\lim_{\theta \to 0} \frac{\sin\left(\pi + \frac{\theta}{2}\right)}{\frac{\theta}{2}}$$

$$= \lim_{\theta \to 0} \frac{-\sin\frac{\theta}{2}}{\frac{\theta}{2}} = -1$$

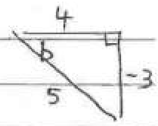
2 Find the exact value of $\sin\left(\cos^{-1}\left(\frac{2}{3}\right) + \tan^{-1}\left(-\frac{3}{4}\right)\right)$. 2

c) $\sin\left(\cos^{-1}\left(\frac{2}{3}\right) + \tan^{-1}\left(-\frac{3}{4}\right)\right) = \sin(a+b)$

Let $\cos^{-1}\frac{2}{3} = a$ $\quad$ $\tan^{-1}\left(\frac{-3}{4}\right) = b$



$\cos a = \frac{2}{3}$ $\quad$ $\tan b = \frac{-3}{4}$



(1st quad) $\quad$ (4th quad)

Now $\sin(a+b) = \sin a \cos b + \cos a \sin b$

$$= \frac{\sqrt{5}}{3} \cdot \frac{4}{5} + \frac{2}{3} \cdot \frac{-3}{5} = \boxed{\frac{4\sqrt{5}-6}{15}}$$

3 a. Prove the trigonometric identity $\tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$. 2

$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(2\theta + \theta) = \frac{\tan 2\theta + \tan\theta}{1 - \tan 2\theta \tan\theta} = \frac{\frac{2\tan\theta}{1-\tan^2\theta} + \frac{\tan\theta}{1}}{1 - \frac{2\tan\theta}{1-\tan^2\theta} \cdot \tan\theta}$$

$$= \frac{\frac{2\tan\theta + \tan\theta - \tan^3\theta}{1-\tan^2\theta}}{\frac{1-\tan^2\theta - 2\tan^2\theta}{1-\tan^2\theta}} = \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$$

as required

b. Use the identity from part (a) and let $x = \tan\theta$, to find the roots of the cubic equation 3

$x^3 - 3x^2 - 3x + 1 = 0$ and hence find the exact value of $\tan\left(\frac{\pi}{12}\right)$.

$x^3 - 3x^2 - 3x + 1 = 0$

$\tan^3\theta - 3\tan^2\theta - 3\tan\theta + 1 = 0$

$1 - 3\tan^2\theta = 3\tan\theta - \tan^3\theta$

$\therefore \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta} = 1$

$\tan 3\theta = 1$

$3\theta = \tan^{-1}(1) + k\pi$ where k is an integer

$\theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{9\pi}{12}\left(\frac{3\pi}{4}\right)$

$\tan\frac{3\pi}{4} = -1$ and so one factor of the cubic is $x+1$

$x^3 - 3x^2 - 3x + 1 = (x+1)(x^2 - 4x + 1)$

$\tan\frac{\pi}{12}$ and $\tan\frac{5\pi}{12}$ are roots of $x^2 - 4x + 1 = 0$

$x = \frac{4 \pm \sqrt{16 - 4(1)(1)}}{2}$

$x = \frac{4 \pm \sqrt{12}}{2} = \frac{4 \pm 2\sqrt{3}}{2} = 2 \pm \sqrt{3}$

Since $\tan\frac{\pi}{12} < \tan\frac{5\pi}{12}$

$\tan\frac{\pi}{12}$ is the smaller root $x = 2 - \sqrt{3}$

4 Find the domain and range of $y = 2\cos^{-1}\left(\frac{x}{2} - 1\right)$. 2

For $y = 2\cos^{-1}\left(\frac{x}{2} - 1\right)$

Range $0 \le y \le 2\pi$ Domain is $D: -1 \le \frac{x}{2} - 1 \le 1$ $0 \le x \le 4$

5 Given $4\sin\theta + 3\cos\theta = -5$, what is the value of $\tan\frac{\theta}{2}$? 2

$t = \tan\frac{\theta}{2} \Rightarrow \sin\theta = \frac{2t}{1+t^2}$ & $\cos\theta = \frac{1-t^2}{1+t^2}$

$4\sin\theta + 3\cos\theta + 5 = 0$

$\Rightarrow \frac{8t}{1+t^2} + \frac{3-3t^2}{1+t^2} + 5 = 0$

$8t + 3 - 3t^2 + 5 + 5t^2 = 0$

$\Rightarrow 2t^2 + 8t + 8 = 0$

$\Rightarrow t^2 + 4t + 4 = 0$

$\Rightarrow (t+2)^2 = 0$

or $t = -2$

$\Rightarrow \tan\frac{\theta}{2} = -2$

6 Which expression is equivalent to $1 - \cos 2x$? 1

A. $\sin^2 x - \cos^2 x$

B. $2(\sin^2 x + 1)$

C. $2\sin^2 x$

D. $2(1 + \cos^2 x)$

$1 - \cos 2x = 1 - (\cos^2 x - \sin^2 x)$
$= \sin^2 x + 1 - \cos^2 x$
$= \sin^2 x + 1 - (1 - \sin^2 x) = 2\sin^2 x$

7 What is the value of $\cos^{-1}(\sin\alpha)$, where $\frac{\pi}{2} < \alpha < \pi$? 1

A. $\pi - \alpha$

B. $\frac{\pi}{2} - \alpha$

C. $\alpha - \frac{\pi}{2}$

D. $\alpha + \frac{\pi}{2}$

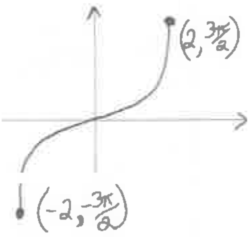
$\cos^{-1}(\sin\alpha)$ $\frac{\pi}{2} < \alpha < \pi$ ie 2nd quad

$\sin\alpha > 0$
$= \frac{a}{c}$
$\therefore \cos^{-1}(\sin\alpha) = \cos^{-1}\left(\frac{a}{c}\right)$
$= \angle ABC$

$\angle BAC = \pi - \alpha$
$\therefore \angle ABC = \frac{\pi}{2} - \angle BAC$
$= \frac{\pi}{2} - (\pi - \alpha)$
$= \alpha - \frac{\pi}{2}$

hence (C)

8 Consider the curve $f(x) = 3\sin^{-1}\left(\frac{x}{2}\right)$. Sketch the curve, clearly indicating the coordinates of any intercepts with the axes and any endpoints. 1



9 Use the relationship $\sin A\cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$, or otherwise, to determine where the curve $y = \sin 5x + \sin x$ cuts the $x-$axis for $0 \le x \le \frac{\pi}{2}$. 2

$\sin 5x + \sin x = 0$

$\sin(3x + 2x) + \sin(3x - 2x) = 0$

$2\sin 3x \cos 2x = 0$

$\sin 3x = 0$ or $\cos 2x = 0$

$3x = 0, \pi, 2\pi, 3\pi, 4\pi$ $2x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots$

$x = 0, \frac{\pi}{3}$, $x = \frac{\pi}{4}$.

10 a. Prove the trigonometric identity $sin3A = 3sinA - 4\,sin^3 A$. 3

LHS $= \sin 3A = \sin(2A + A)$

$= \sin 2A \cos A + \cos 2A \sin A = 2\sin A \cos A \cos A + (1 - 2\sin^2 A)\sin A$

$= 2\sin A \cos^2 A + \sin A - 2\sin^3 A = 2\sin A(1 - \sin^2 A) + \sin A - 2\sin^3 A$

$= 2\sin A - 2\sin^3 A + \sin A - 2\sin^3 A = 3\sin A - 4\sin^3 A =$ RHS

$\therefore \sin 3A = 3\sin A - 4\sin^3 A$

b. Hence, show that the equation $6x - 8x^3 = 1$ has the roots $sin\frac{\pi}{18}, sin\frac{5\pi}{18}$ and $sin\frac{25\pi}{18}$. 3

Hint: Let $x = sinA$.

Hence show $6x^3 - 8x^3 = 1$ has roots

$\sin\frac{\pi}{18}, \sin\frac{5\pi}{18}, \sin\frac{25\pi}{18}$

a cubic has at most 3 real solutions

So if we find 3 distinct solutions

we have found them all (as degree 3)

let $x = \sin A$ — ②

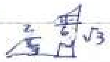
sub ② in ① $6\sin A - 8\sin A = 1$

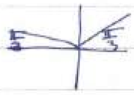
$3\sin A - 4\sin A = \frac{1}{2}$

$\therefore$ using part (a) $\sin 3A = 3\sin A - 4\sin^3 A$.

$\sin 3A = \frac{1}{2}$.

Sine is positive in Q1 & Q2





$\sin\frac{\pi}{6} = \frac{1}{2}$

$\therefore R = \frac{\pi}{6}$

$\therefore 3A = \frac{\pi}{6} + 2n(\pi)$ or $3A = (\pi - \frac{\pi}{6}) + 2n\pi$.

$A = \frac{\pi}{18} + \frac{2n\pi}{3}$ or $A = \frac{5\pi}{18} + \frac{2n\pi}{3}$

$\therefore A = \frac{\pi}{18}, \frac{5\pi}{18}, \frac{13\pi}{18}, \frac{17\pi}{18}, \ldots, \frac{19\pi}{18}, \ldots \frac{25\pi}{18}, \ldots$

There will be repeats but only 3 distinct solutions...

$\therefore$ the solutions are.

$\sin\frac{\pi}{18}$, $\sin\frac{5\pi}{18}$, and $\sin\frac{25\pi}{18}$,

c. Hence show that $sin\frac{\pi}{18} \times sin\frac{5\pi}{18} \times sin\frac{25\pi}{18} = -\frac{1}{8}$. 1

Product of roots: $sin\frac{\pi}{18} \times sin\frac{5\pi}{18} \times sin\frac{25\pi}{18} = -\frac{d}{a} = -\frac{1}{8}$

11 Solve the simultaneous equations for x and y leaving your answer in simplest exact form: 2

$$2\,sin^{-1} x + 3\,cos^{-1} y = \frac{3\pi}{2}$$

$$2\,sin^{-1} x - 3\,cos^{-1} y = \frac{\pi}{2}$$

let $\alpha = \sin^{-1}(x)$ $\quad \beta = \cos^{-1} y$

$2\alpha + 3\beta = \frac{3\pi}{2}$ $\quad 2\alpha - 3\beta = \frac{\pi}{2}$

$4\alpha = 2\pi \Rightarrow \alpha = \frac{\pi}{2}$

as $-\frac{\pi}{2} \le \sin^{-1}\theta \le \frac{\pi}{2}$

$6\beta = \pi \Rightarrow \beta = \frac{\pi}{6}$

as $0 \le \cos^{-1}\theta \le \pi$

$\frac{\pi}{2} = \sin^{-1}(x)$

$x = 1$

$\frac{\pi}{6} = \cos^{-1} y$

$y = \frac{\sqrt{3}}{2}$

13 a. Show that $cos3x = 4\,cos^3 x - 3cosx$. 2

$\cos 3x = \cos(2x + x)$

$= \cos 2x \cos x - \sin 2x \sin x$

$= (2\cos^2 x - 1)\cos x - 2\sin x \cos x \sin x$

$= 2\cos^3 x - \cos x - 2\sin^2 x \cos x$

b. Find all solutions to the equation $cos3x + 2cosx = 0$ over the domain $[-2\pi, 2\pi]$. 2

$$\cos 3x + 2\cos x = 0$$
$$4\cos^3 x - 3\cos x + 2\cos x$$
$$\cos x(4\cos^2 x - 1) = 0$$
$$\cos x = 0$$
$$x = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}$$

$$\cos^2 x = \frac{1}{4}$$
$$\cos x = \pm\frac{1}{2}$$
$$x = \pm\left(\frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}\right)$$

Further Trigonometry
Topic Test 3 Solution

Total Marks: 30

Time: 50 minutes

1 a. Show that $tan\left(\theta+\frac{\pi}{6}\right)=\frac{1+\sqrt{3}tan\theta}{\sqrt{3}-tan\theta}$. 1

$$\tan\left(\theta+\frac{\pi}{6}\right)=\frac{\tan\theta+\tan\frac{\pi}{6}}{1-\tan\theta\tan\frac{\pi}{6}}=\frac{\tan\theta+\frac{1}{\sqrt{3}}}{1-\frac{\tan\theta}{\sqrt{3}}}=\frac{1+\sqrt{3}\tan\theta}{\sqrt{3}-\tan\theta}$$

b. Hence, or otherwise, solve for $0\le\theta\le\pi$, $1+\sqrt{3}tan\theta=(\sqrt{3}-tan\theta)\,tan(\pi-\theta)$ 2

$1+\sqrt{3}\tan\theta=(\sqrt{3}-\tan\theta)\tan(\pi-\theta)$

$\therefore \frac{1+\sqrt{3}\tan\theta}{\sqrt{3}-\tan\theta}=\tan(\pi-\theta)$

$\tan\left(\theta+\frac{\pi}{6}\right)=\tan(\pi-\theta)$ (from i)

$\theta+\frac{\pi}{6}=\pi-\theta$ or $\theta+\frac{\pi}{6}=\pi+(\pi-\theta)$

$2\theta=\pi-\frac{\pi}{6}$ or $2\theta=2\pi-\frac{\pi}{6}$

$2\theta=\frac{5\pi}{6}$ or $2\theta=\frac{11\pi}{6}$

$\theta=\frac{5\pi}{12}$ or $\frac{11\pi}{12}$

2 a. Show that $sin2\theta+sin4\theta-sin6\theta=4sin3\theta sin2\theta sin\theta$. 2

We can aim to use sums to products:

$\sin 2\theta+\sin 4\theta-\sin 6\theta$

$=\sin(3\theta-\theta)+\sin(3\theta+\theta)-\sin 6\theta$

$=2\sin 3\theta\cos\theta-\sin 6\theta$ sum to product

$=2\sin 3\theta\cos\theta-2\sin 3\theta\cos 3\theta$ double angle

$=2\sin 3\theta(\cos\theta-\cos 3\theta)$

$=2\sin 3\theta(\cos(2\theta-\theta)-\cos(2\theta+\theta))$

$=2\sin 3\theta(2\sin 2\theta\sin\theta)$ sum to product

$=4\sin 3\theta\sin 2\theta\sin\theta$

b. Hence, solve $sin2\theta+sin4\theta=sin6\theta$ for $0\le\theta\le\pi$. 2

$\sin 2\theta+\sin 4\theta=\sin 6\theta$

$\sin 2\theta+\sin 4\theta-\sin 6\theta=0$

$4\sin 3\theta\sin 2\theta\sin\theta=0$ by (i)

Hence either:

$\sin\theta=0, \sin 2\theta=0, \sin 3\theta=0$

If $\sin\theta=0$, then $\theta=0,\pi$

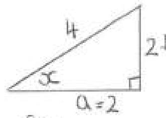
If $\sin 2\theta=0$, then $\theta=0,\frac{\pi}{2},\pi$

If $\sin 3\theta=0$, then $\theta=0,\frac{\pi}{3},\frac{2\pi}{3},\pi$

Hence the solution set is $\theta=0,\frac{\pi}{3},\frac{\pi}{2},\frac{2\pi}{3},\pi$

3 What is the value of $sin2x$ given that $sinx = \frac{2\sqrt{3}}{4}$ and x is obtuse? 1

$\sin x = \frac{2\sqrt{3}}{4}$



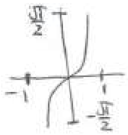
$a^2 = 4^2 - (2\sqrt{3})^2$
$= 16 - 12$
$= 4 \quad a = 2$

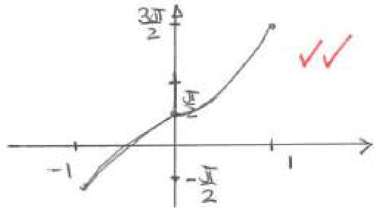
given: x is obtuse

then $\sin 2x = 2\sin x \cos x$
$= 2 \times \frac{2\sqrt{3}}{4} \times \frac{(-)2}{4} = -\frac{\sqrt{3}}{2}$

4 Sketch $y = 2\,sin^{-1}\,x + \frac{\pi}{2}$ for $[-1, 1]$. 2

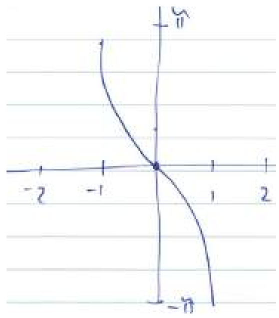
Sketch by 2 : $\frac{\pi}{2} \times 2 = \pi$
and $-\frac{\pi}{2} \times 2 = -\pi$
↑ by $\frac{\pi}{2}$: $\pi + \frac{\pi}{2} = \frac{3\pi}{2}$ and $-\pi + \frac{\pi}{2} = -\frac{\pi}{2}$





✓✓

5 Sketch the graph of $y = 2\,sin^{-1}(-x)$. 2



6 Consider the functions $f(x) = cos^{-1}\,x$ and $g(x) = 2\,tan^{-1}(1 - x)$. Show that graphs of the functions $f(x)$ and $g(x)$ intersect each other on the y-axis at $y = \frac{\pi}{2}$. 1

$f(x) = \cos^{-1}(x)$
$f(0) = \cos^{-1}(0)$
$= \frac{\pi}{2}$

$g(x) = 2\tan^{-1}(1-x)$
$g(0) = 2\tan^{-1}(1)$
$= 2\left(\frac{\pi}{4}\right) = \frac{\pi}{2}$

Since $f(0) = g(0)$, they intersect the y-axis at the same point.

7 Solve $sin7x + sinx = sin4x$ for $0 \le x \le \frac{\pi}{2}$. 3

now $\frac{1}{2}(\sin(4x+3x) + \sin(4x-3x)) = \sin 4x \cos 3x$
$\therefore \sin 7x + \sin x = 2\sin 4x \cos 3x$
and $\sin 4x = 2\sin 4x \cos 3x$
$0 = 2\sin 4x \cos 3x - \sin 4x$
$0 = \sin 4x(2\cos 3x - 1)$
$\therefore \sin 4x = 0$ or $2\cos 3x - 1 = 0$
$4x = 0, \pi, 2\pi, 3\pi, 4\pi$ $\qquad \cos 3x = \frac{1}{2}$
$\therefore x = 0, \frac{\pi}{4}, \frac{\pi}{2}, \cancel{\frac{3\pi}{4}}, \cancel{\pi}$ ✓ $\qquad 3x = \frac{\pi}{3}, 2\pi - \frac{\pi}{3}$
$3x = \frac{\pi}{3}, \frac{5\pi}{3}$
$x = \frac{\pi}{9}, \cancel{\frac{5\pi}{9}}$

8 a. Prove the trigonometric identity $tan3\theta = \frac{3tan\theta - tan^3\theta}{1-3\,tan^2\theta}$. 3

$$\tan 3\theta = \tan(2\theta+\theta) = \frac{\tan 2\theta + \tan\theta}{1-\tan 2\theta \tan\theta} = \frac{\left(\frac{2\tan\theta}{1-\tan^2\theta}\right)+\tan\theta}{1-\left(\frac{2\tan\theta}{1-\tan^2\theta}\right)\tan\theta}$$

$$= \frac{\frac{2\tan\theta + \tan\theta(1-\tan^2\theta)}{1-\tan^2\theta}}{\frac{1-\tan^2\theta - 2\tan^2\theta}{1-\tan^2\theta}} = \frac{2\tan\theta + \tan\theta - \tan^3\theta}{1-3\tan^2\theta}$$

$$= \frac{3\tan\theta - \tan^3\theta}{1-3\tan^2\theta}$$

b. Hence show that the eqation $x^3 - 3\sqrt{3}x^2 - 3x + \sqrt{3} = 0$ has roots $tan\frac{\pi}{9}, tan\frac{4\pi}{9}$ and $tan\frac{7\pi}{9}$. 2

$x^3 - 3\sqrt{3}x^2 - 3x + \sqrt{3} = 0$ rearrange to look like (i).

$\sqrt{3} - 3\sqrt{3}x^2 = 3x - x^3$

$\sqrt{3}(1-3x^2) = 3x - x^3$

$\sqrt{3} = \frac{3x - x^3}{1-3x^2}$ let so $x = \tan\theta$

$\therefore \tan 3\theta = \sqrt{3}$

$3\theta = \frac{\pi}{3}, \frac{4\pi}{3}, \frac{7\pi}{3}, \ldots$

$\theta = \frac{\pi}{9}, \frac{4\pi}{9}, \frac{7\pi}{9}$ $x = \tan\frac{\pi}{9}, \tan\frac{4\pi}{9}, \frac{7\pi}{9}$

c. Hence, without the use of a calculator, show that $tan^2\frac{\pi}{9}, + tan^2\frac{4\pi}{9} + tan^2\frac{7\pi}{9} = 33$. 2

So if roots are α, β, λ then $\alpha^2+\beta^2+\lambda^2 = (\alpha+\beta+\lambda)^2 - 2(\alpha\beta+\beta\lambda + \lambda\alpha)$

$\alpha+\beta+\lambda = \frac{-b}{a} = \frac{3\sqrt{3}}{1}$ $\alpha\beta+\beta\lambda+\lambda\alpha = \frac{c}{a} = \frac{-3}{1}$

$\therefore = (3\sqrt{3})^2 - 2(-3)$

$= 9(3) + 6$

$= 33.$

9 What is the domain and range of $y = cos^{-1}\left(\frac{3x}{2}\right)$? 1

Domain:

$-1 \le \frac{3x}{2} \le -1$

$-2 \le 3x \le 2$

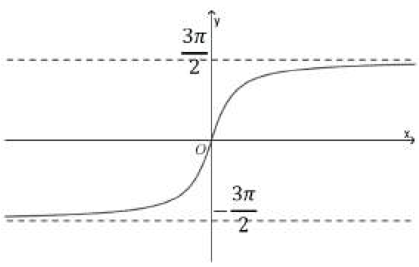
$-\frac{2}{3} \le x \le \frac{2}{3}$

Range:

$0 \le \cos^{-1}\left(\frac{3x}{2}\right) \le \pi$

$0 \le y \le \pi$

10 Sketch the graph of $y = 3\,tan^{-1}x$. 2



11 Given that $t = tan\frac{\theta}{2}$, write $tan\theta sin\theta$ in terms of t. 1

$t = \tan\frac{\theta}{2}$; $\tan\theta = \frac{2t}{1-t^2}$, $\sin\theta = \frac{2t}{1+t^2}$

$\therefore \tan\theta\sin\theta = \frac{2t}{1-t^2}\cdot\frac{2t}{1+t^2} = \frac{4t^2}{1-t^4}$

12 Use the substitution $t = tan\frac{x}{2}$ to show that $secx + tanx = tan\left(\frac{\pi}{4} + \frac{x}{2}\right)$. 3

$$t = \tan\frac{x}{2}$$
$$LHS = \sec x + \tan x$$
$$LHS = \frac{1+t^2}{1-t^2} + \frac{2t}{1-t^2} = \frac{t^2+2t+1}{1-t^2}$$
$$= \frac{(t+1)^2}{(1+t)(1-t)} = \frac{1+t}{1-t} = \frac{\tan\frac{\pi}{4} + \tan\frac{x}{2}}{1 - \tan\frac{\pi}{4}\tan\frac{x}{2}}$$
$$= \tan\left(\frac{\pi}{4} + \frac{x}{2}\right) = RHS$$